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AHEAD Online

JDBC Introduction

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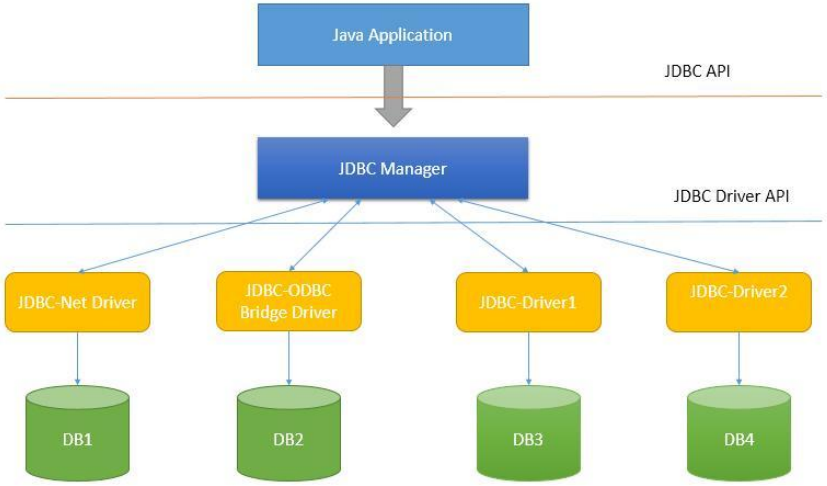
Objectives

- JDBC.

What is JDBC

- JDBC stands for JavaDataBaseConnectivity.
- It is an Application Programming Interface(API) used to connect Java application with Database.



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What is JDBC

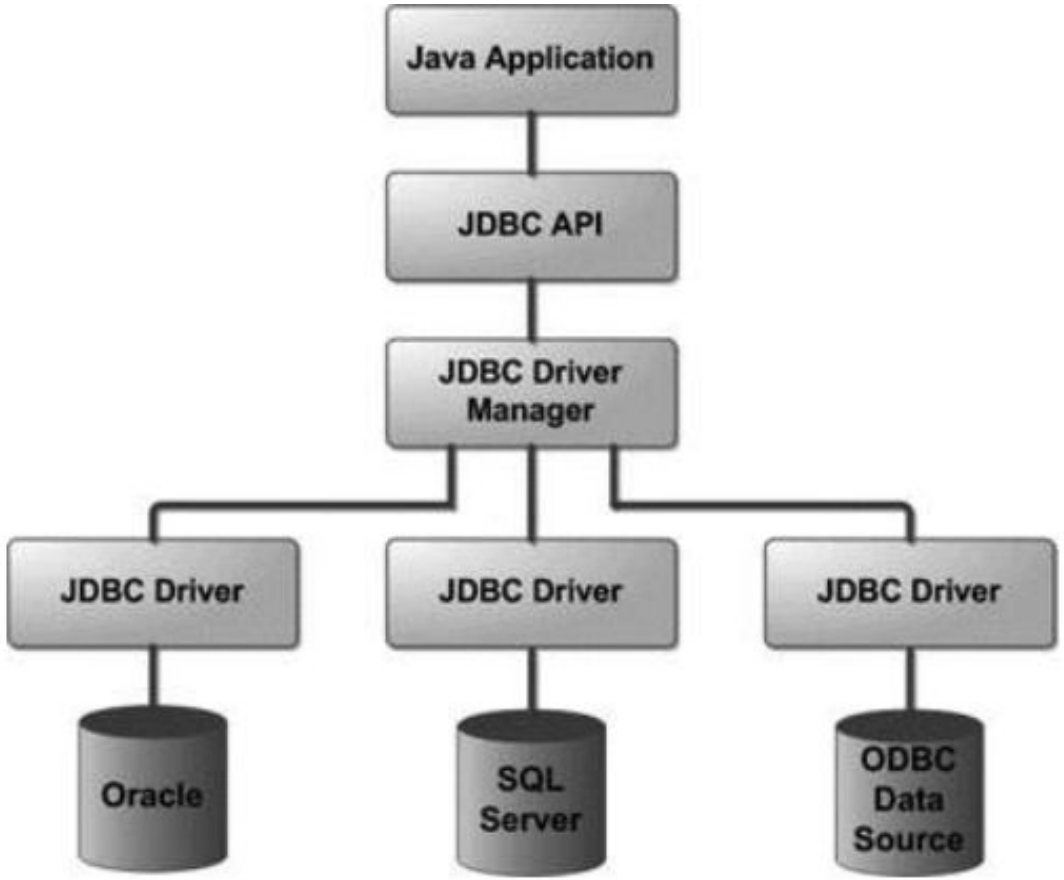
- JDBC is used to interact with various type of Database such as Oracle, MS Access, My SQL and SQL Server.
- It basically acts as a channel between your Java program and databases.
- We can use JDBC API to handle database using Java program and can perform the following activities:

1.Connect to the database 2.Execute queries and update statements to the database 3.Retrieve the result received from the database.

JDBC Architecture

- The JDBC API supports both two-tier and three-tier processing models for database access but in general, JDBC Architecture consists of two layers
- JDBC API: This provides the application-to-JDBC Manager connection.
- JDBC Driver API: This supports the JDBC Manager-to-Database Connection.

JDBC Architecture



JDBC Components

- DriverManager: Matches connection requests from the java application with the proper database driver using communication sub protocol.
- Driver: This interface handles the communications with the database server.
- Connection: This interface has methods for contacting a database. For submitting the SQL statements to the database.

JDBC Components

- ResultSet: These objects hold data retrieved from a database after you execute an SQL query using Statement objects.
- SQLException: This class handles any errors that occur in a database application.

JDBC Driver Manager

- Driver Manager helps to connect an application based on the database connection string.
- In JDBC 4.0,the JDBC drivers are automatically loaded based on the classpath.
- Legacy JDBC 3.0 drivers have to be explicitly loaded with Class.forName(the driver Name)

JDBC Driver Models

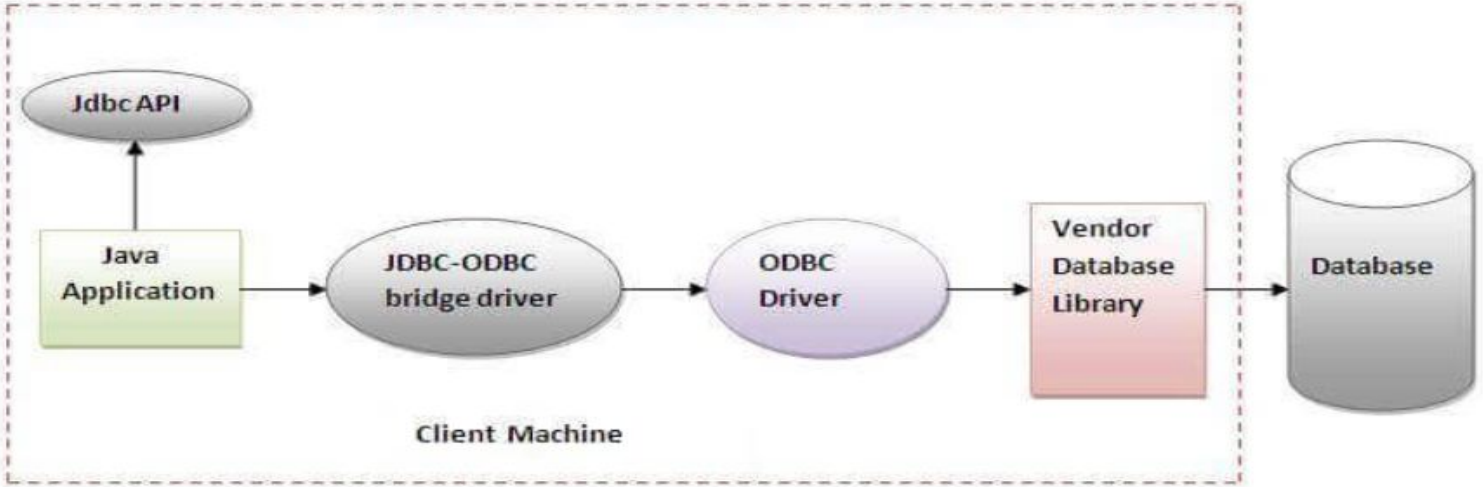
- JDBC Driver is a software component that enables java application to interact with the database.

There are 4 types of JDBC drivers:

- JDBC-ODBC bridge driver
- Native-API driver (partially java driver)
- Network Protocol driver (fully java driver)
- Thin driver (fully java driver)

JDBC ODBC Driver

- The JDBC-ODBC bridge driver uses ODBC driver to connect to the database. The JDBC-ODBC bridge driver converts JDBC method calls into the ODBC function calls.



JDBC ODBC bridge driver

Advantages:

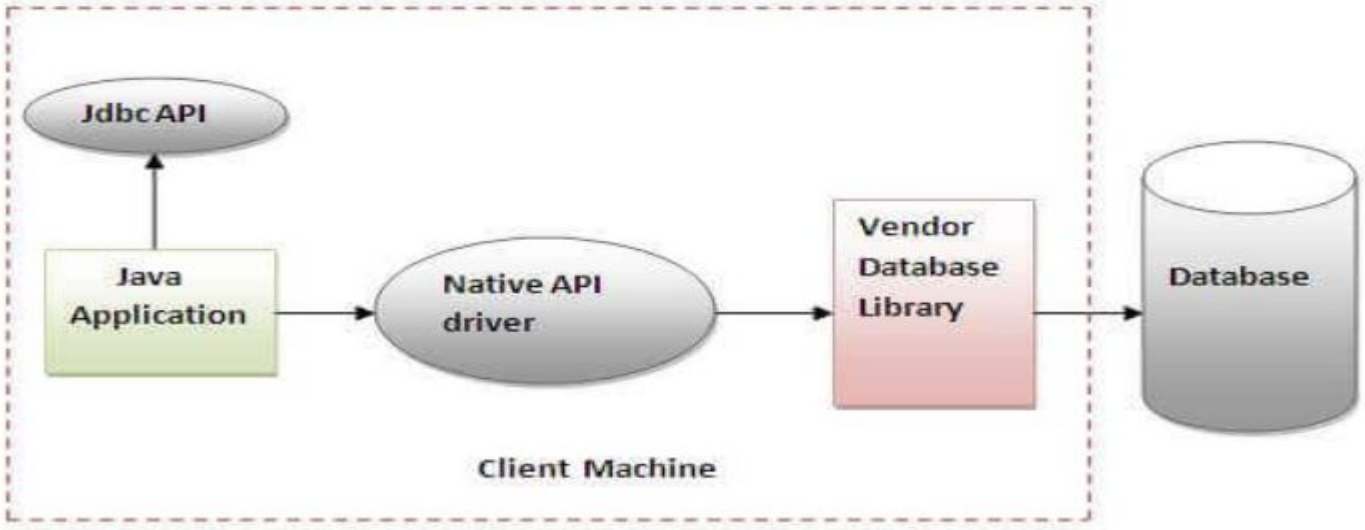
- easy to use.
- can be easily connected to any database.

Disadvantages:

- Performance degraded because JDBC method call is converted into the ODBC function calls.
- The ODBC driver needs to be installed on the client machine.

Native API driver

- The Native API driver uses the client-side libraries of the database. The driver converts JDBC method calls into native calls of the database API.



Native API driver

Advantage:

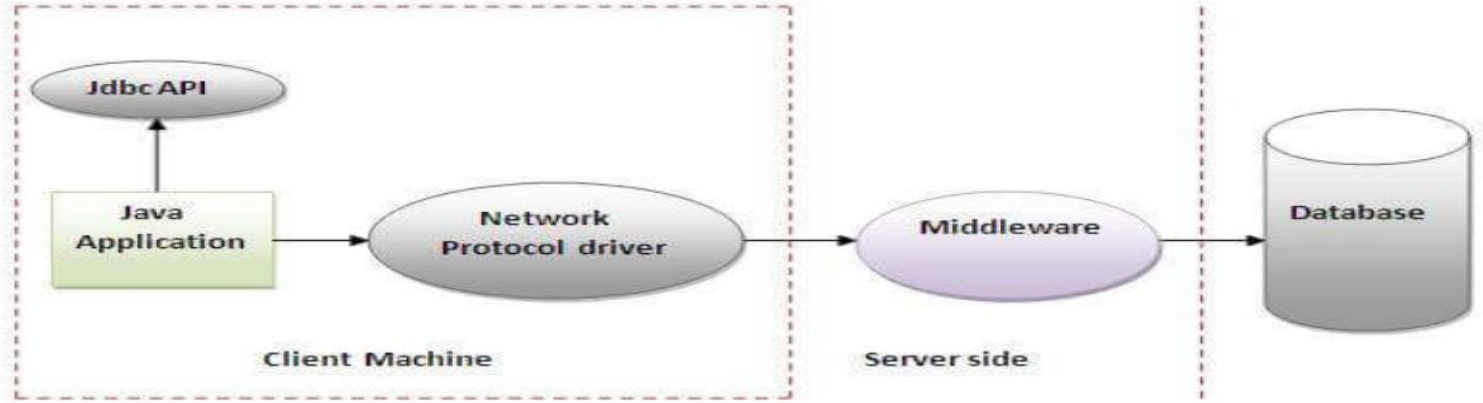
- performance upgraded than JDBC-ODBC bridge driver.

Disadvantage:

- The Native driver needs to be installed on the each client machine.
- The Vendor client library needs to be installed on client machine.

Network Protocol Driver

- The Network Protocol driver uses middleware (application server) that converts JDBC calls directly or indirectly into the vendor-specific database protocol.



Network Protocol Driver

Advantage:

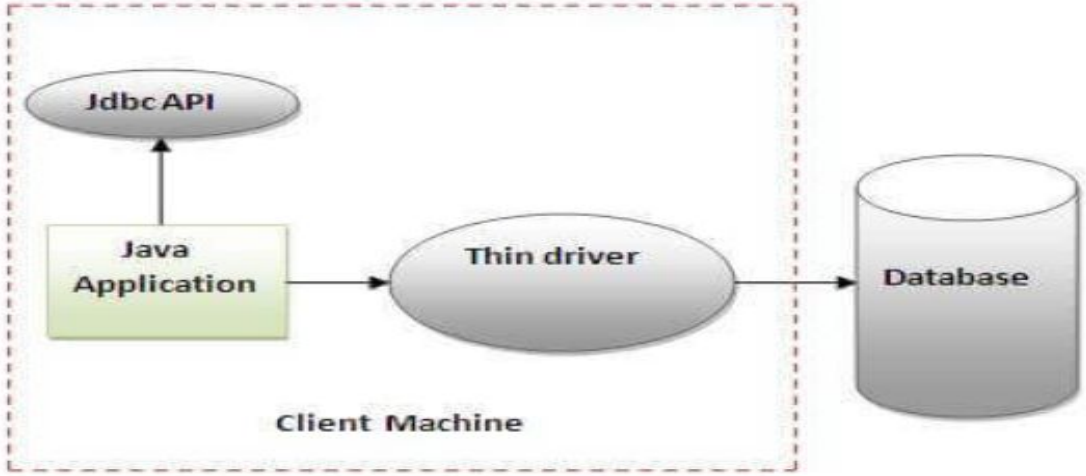
- No client side library is required because of application server that can perform many tasks like auditing, load balancing, logging etc.

Disadvantages:

- Network support is required on client machine.
- Requires database-specific coding to be done in the middle tier.

Thin Driver

- The thin driver converts JDBC calls directly into the vendor-specific database protocol. That is why it is known as thin driver. It is fully written in Java language.



Thin Driver

Advantage:

- Better performance than all other drivers.
- No software is required at client side or server side.

Disadvantage:

- Drivers depend on the Database.

Takeaways

- JDBC

Reference

[1] Naughton P. and Schildt H., “Java2 Complete Reference”, Eighth Edition, Tata McGraw- Hill, 2011.